

Module 07: Flow Control Foundations & Stop-and-Wait ARQ Protocol

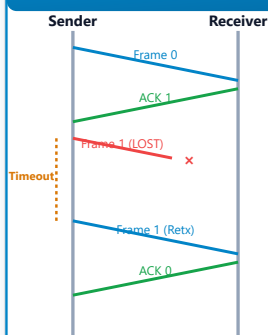
Unit 2: Data Link Layer & MAC Sublayer

Flow Control & Error Control

Rigorous study of Stop-and-Wait ARQ protocol mechanics, timer timeouts, 1-bit sequence numbering, mathematical efficiency derivation, and satellite link numericals.

Stop-and-Wait ARQ: Protocol Timing, Failure Recovery & Efficiency Derivation

1. Protocol Operation & Failure Scenarios



• Key Protocol Rules:

1. Sender sends 1 frame, starts Timer.
2. Receiver accepts, sends ACK for next expected frame (ACK 1 expects Frame 1).
3. If timer expires: Retransmit frame.

• Why 1-bit Sequence Numbers?

Handles Duplicate Frames when ACK is lost!
Sequence numbers toggle: 0 and 1.
Window Size Sender = 1, Receiver = 1.

2. Efficiency & Utilization Derivation

$$\text{Total Cycle Time} = T_t + 2 \cdot T_p$$

(Assuming ACK transmission time & processing delay are negligible)

$$\eta = T_t / (T_t + 2T_p) = 1 / (1 + 2a)$$

where $a = T_p / T_t$

Throughput / Useful Bandwidth:

$$\text{Throughput} = \eta \cdot \text{Bandwidth} = L / (T_t + 2T_p)$$

For large T_p (Satellite links, $a \gg 1$): η drops below 1%!

AKTU Core Numerical: Stop-and-Wait ARQ on Satellite Links (Impact of Parameter 'a')

Problem Data: Bandwidth $B = 1 \text{ Mbps}$ (10^6 bps), Frame Length $L = 1000 \text{ bits}$, Propagation Delay $T_p = 250 \text{ ms}$ (0.25 s).

Step 1 (T_t): $T_t = L / B = 1000 / 10^6 = 1 \text{ ms}$ (0.001 s)

Step 2 (Ratio a): $a = T_p / T_t = 250 \text{ ms} / 1 \text{ ms} = 250$

Step 3 (Efficiency): $\eta = 1 / (1 + 2 \cdot 250) = 1 / 501 \approx 0.00199$ (0.199%) | **Throughput** = $0.00199 \times 1 \text{ Mbps} = 1.99 \text{ kbps}$

Sender spends 99.8% of time sitting idle waiting for ACK! This motivates Sliding Window Pipelining.

1. Stop-and-Wait ARQ Protocol Mechanics

Stop-and-Wait ARQ enforces strict sequential flow control: the transmitter sends exactly one frame and starts a timeout timer. It pauses transmission until a valid Acknowledgment (ACK) arrives. 1-bit alternating sequence numbering (0 and 1) prevents duplicate frame delivery caused by lost ACKs.

2. Efficiency Derivation: The $1/(1+2a)$ Law

Fundamental Stop-and-Wait Efficiency Formula

$$a = T_p / T_t = (d / v) / (L / B)$$

$$\eta = T_t / (T_t + 2T_p) = 1 / (1 + 2a)$$

$$\text{Throughput} = \eta \cdot \text{Bandwidth}$$

For high bandwidth-delay product links (such as satellite communication where $a = 250$), link efficiency collapses to less than 0.2%, compelling the development of pipelined Sliding Window protocols.